Mohamed Rayan S P

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years of experience delivering production-grade enterprise AI solutions for the retail sector across GCP and AWS. Core expertise spans multi-agent systems, advanced RAG workflows and cloud-native automation focused on system reliability and measurable business impact.

PROFESSIONAL EXPERIENCE

TATA CONSULTANCY SERVICES 08/2024 – Present | Chennai, India

AI Engineer

Supply Chain Document Processing Automation

- Deployed a serverless GCP-based document processing pipeline to automate auto parts vendor packing slip extraction and JDE ERP integration across 20+ suppliers, projecting savings of 2,800–3,700 manual hours per distribution center annually.
- Boosted extraction accuracy from 75% to 95% by transitioning to a generative AI schema framework, achieving 80% end-to-end automation across non-standard supplier templates.

Multi-agent Insurance Claims Processing System

- Architected a LangGraph-based multi agent insurance claims pipeline with interrupt checkpoints for human review of high-risk claims, backed by deterministic guardrails, PII masking and audit logs.
- Engineered an automated CI test suite and custom evaluation engine combining full branch-coverage graph checks, hand-calibrated LLM-as-a-judge scoring, and trajectory path-conformance checks.

Enterprise IT ServiceDesk Assistant

- Developed an enterprise IT ticket triage RAG assistant for support staff, enforcing server-side pre-retrieval access control for internal SOPs and confidence-gated fallback synthesis.
- Designed a hybrid BM25 + dense search and cross-encoder reranking engine, achieving 95% Recall@5 and 0.81 NDCG@5, validated via prompt-injection testing, dual-layer citation checks, and LLM-as-a-judge evals.

Virtual TryOn Integration – Apparel Commerce

- Led a 5-member team, delivering a Bedrock-powered Virtual Try-On feature across 21 apparel categories in a Dynamic Yield environment on AWS, driving shopper engagement with projected 5% conversion uplift and 10% return-rate reduction.
- Implemented enterprise AI governance and security using AWS Bedrock Guardrails to enforce automated content moderation, safe image filtering and data privacy compliance for user-uploaded photos.

TECHNICAL SKILLS

Generative AI: LLMs, Vector databases, RAG, Agentic Workflows, MCP, Prompt Engineering, NLP, Machine Learning
Frameworks and Tools: PyTorch, Langchain, LangGraph, LangSmith, OpenTelemetry, Copilot Studio, Claude Code
Backend: Python, FastAPI, REST APIs, SQL, MongoDB, Flask, FAISS, ChromaDB, Qdrant
Cloud Technologies & DevOps: AWS, Google Cloud Platform, Docker, Git

CERTIFICATIONS

- Claude Certified Architect - Professional | Anthropic | 2026
- Claude Certified Architect - Foundations | Anthropic | 2026
- Claude Certified Developer - Foundations | Anthropic | 2026

EDUCATION

B.Tech in Computer Science and Engineering 2020 – 2024 | Vellore, India

Vellore Institute of Technology